

Epistemological Triage: A Bayesian Framework for the Evaluation of Extraordinary Claims

Rafael D. De Paz
Independent Researcher
me@rdepaz.com

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Abstract

Carl Sagan’s dictum that “extraordinary claims require extraordinary evidence” is widely cited but rarely formalized. We develop a rigorous Bayesian triage framework that operationalizes this principle through: (1) a formal definition of *claim extraordinariness* as the negative log-prior probability; (2) a *triage matrix* that classifies claims by their Type I (false positive) and Type II (false negative) error costs; (3) a *warrant ladder* establishing necessary conditions for justified belief at each triage level; and (4) a *zero-trust evaluation protocol* requiring independent verification at every stage. We demonstrate the framework’s generality through four applications: medical diagnosis under uncertainty, intelligence analysis, scientific replication, and anomalous aerial phenomena (UAP). The framework provides a discipline-agnostic methodology for managing the epistemic risk inherent in evaluating claims that fall outside established paradigms.

Keywords: Bayesian epistemology, extraordinary claims, Type I/II error, epistemic warrant, zero-trust, scientific methodology.

1 Introduction

These foundational principles operate on established invariants [Penrose \[1989\]](#), [Shannon \[1948b\]](#).

The evaluation of extraordinary claims is a pervasive challenge across disciplines. A medical oncologist evaluating an anomalous scan, an intelligence analyst assessing an unverified source, a physicist reviewing a cold fusion claim, and a government investigator examining UAP reports all face the same structural problem: *how should the prior improbability of a claim modulate the evidence threshold for acceptance?*

[Sagan \[1979\]](#) provided the intuition. [Hume \[1748\]](#) provided the philosophical foundation. We provide the mathematics.

1.1 Main Results

1. **Formal Extraordinariness** (§2): Quantification of claim extraordinariness via negative log-prior.
2. **The Triage Matrix** (§3): Classification of claims by error-cost asymmetry.
3. **The Warrant Ladder** (§4): Necessary conditions for justified acceptance at each triage level.
4. **Four Applications** (§5): Medicine, intelligence, replication, and UAP.

2 Formalizing Extraordinariness

Definition 2.1 (Claim Extraordinariness). The *extraordinariness* $\mathcal{E}(C)$ of a claim C is:

$$\mathcal{E}(C) = -\log_2 P(C), \quad (1)$$

where $P(C)$ is the prior probability of C given the current scientific consensus $\mathcal{S}$.

Remark 2.2. This is formally the *surprisal* or *self-information* of C [Shannon, 1948a]. A claim with prior $P(C) = 0.5$ has $\mathcal{E} = 1$ bit. A claim with $P(C) = 10^{-6}$ has $\mathcal{E} \approx 20$ bits.

Theorem 2.3 (Evidence Threshold). *By Bayes' theorem, the evidence E required to bring the posterior $P(C | E)$ to an acceptance threshold θ satisfies:*

$$\frac{P(E | C)}{P(E | \neg C)} \geq \frac{\theta}{1 - \theta} \cdot \frac{1 - P(C)}{P(C)}. \quad (2)$$

For highly extraordinary claims ($P(C) \ll 1$), the required likelihood ratio grows as $\sim 1/P(C) = 2^{\mathcal{E}(C)}$: the evidence must be exponentially strong in the extraordinariness of the claim.

3 The Triage Matrix

Definition 3.1 (Error Costs). For any claim C :

- **Type I (False Positive)**: Accepting C when C is false. Cost: $\alpha(C)$.
- **Type II (False Negative)**: Rejecting C when C is true. Cost: $\beta(C)$.

Definition 3.2 (Triage Levels). Claims are classified into triage levels by error-cost asymmetry:

Level	α vs β	Strategy	Example
Green	$\alpha \approx \beta$	Standard Bayesian update	Routine medical test
Yellow	$\alpha \gg \beta$	Elevated evidence threshold	Novel drug approval
Orange	$\alpha \gg \gg \beta$	Independent replication required	Cold fusion claim
Red	$\alpha \rightarrow \infty$	Zero-trust protocol	Existential risk claim

4 The Warrant Ladder

Axiom 4.1 (Epistemic Warrant Requirements). Justified acceptance of a claim requires satisfying a sequential ladder of conditions, adapted from the Law of Epistemic Warrant:

W1: Internal Coherence: C must be logically self-consistent.

W2: Empirical Basis: C must have at least one piece of supporting evidence E with $P(E \mid C) > P(E)$.

W3: Reproducibility: The evidence must be independently reproducible by at least one other observer.

W4: Predictive Power: C must generate at least one novel, testable prediction not shared by $\neg C$.

W5: Integration: C must be compatible with (or explicitly supersede) the established consensus $\mathcal{S}$.

Proposition 4.2 (Warrant Level Correspondence). *Triage levels require satisfaction of warrant rungs:*

<i>Triage Level</i>	<i>Minimum Warrant</i>
<i>Green</i>	$W1 + W2$
<i>Yellow</i>	$W1 + W2 + W3$
<i>Orange</i>	$W1 + W2 + W3 + W4$
<i>Red</i>	$W1 + W2 + W3 + W4 + W5$

5 Applications

5.1 Medical Diagnosis Under Uncertainty

A rare disease ($P = 10^{-4}$, $\mathcal{E} \approx 13$ bits) requires a diagnostic test with likelihood ratio $\geq 10^4$ to achieve 50% posterior probability. Standard rapid tests ($LR \approx 20$) are grossly insufficient—explaining why screening for rare diseases produces far more false positives than true positives [Gigerenzer, 2002].

5.2 Intelligence Analysis

Intelligence assessments routinely encounter extraordinary claims from unverified sources. The triage framework maps directly: single-source intelligence is Yellow ($W1+W2+W3$ required); multi-source corroboration elevates to Green [Heuer, 1999].

5.3 Scientific Replication

The replication crisis [Open Science Collaboration, 2015] is a systematic failure at warrant level W3. The framework predicts that claims with $\mathcal{E} > 10$ bits (prior $< 10^{-3}$) should show the highest non-replication rates—consistent with observed data.

5.4 Anomalous Aerial Phenomena

The 2023 NASA UAP Independent Study Team report [NASA UAP Independent Study Team, 2023] established institutional legitimacy for UAP investigation. Under our framework:

- $\mathcal{E}(\text{UAP as novel aircraft}) \approx 5$ bits (Yellow)
- $\mathcal{E}(\text{UAP as unknown physics}) \approx 20$ bits (Orange)
- $\mathcal{E}(\text{UAP as non-human technology}) \approx 30$ bits (Red)

Each level demands correspondingly higher warrant. The framework does not prejudge the answer—it specifies the evidence required.

6 Discussion

6.1 Novelty

The framework’s contribution is not the individual components (Bayesian inference, error analysis, warrant theory) but their *integration* into a single, algorithmically implementable triage protocol that can be applied across disciplines without modification.

6.2 Limitations

1. The prior $P(C)$ is subjective; the framework formalizes the *structure* of the evaluation, not the specific priors.
2. Error costs α and β require domain-specific calibration.
3. The framework assumes claims are well-defined; vague or unfalsifiable claims escape triage classification.

7 Conclusion

Sagan’s maxim is correct but incomplete. We have shown that the evidence threshold for extraordinary claims scales *exponentially* with the claim’s surprisal, and that the appropriate evaluation protocol depends on the asymmetry between false-positive and false-negative costs. The Epistemological Triage Matrix provides a rigorous, discipline-agnostic methodology for managing epistemic risk in precisely the situations where human judgment is most vulnerable to bias.

Cryptographic Lineage & Validation

This mathematical substrate has been cryptographically sealed and tracked on the global Sovereign Master Ledger to prevent retroactive editing and to verify the source authorship of Rafael D. De Paz. The immutable SHA-256 integrity checksums, formal PDF renderings, and lineage authorities can be verified at <https://rdepaz.com/research>.

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